

Q1	Suggested Answer
(a)(i)	Legend: N – normal contact force T – tension of string W_C – weight of block A Forces correctly labelled (ruler should be used) Correct direction indicated (T longer in length than component of W_C along slope)
(a)(ii)	Taking mass B as the system $m_B g - T = m_B a$ $(8.3 \times 9.81) - 54 = 8.3a$ $a = 3.30 \text{ m s}^{-2}$ Taking mass A as the system $T - m_A g \sin 50^\circ = m_A a$ $54 - (m_A \times 9.81 \sin 50^\circ) = m_A \times 3.30$ $m_A = 4.99 \text{ kg}$ $= 5.0 \text{ kg (2 s.f.) (shown)}$
(b)(i)	Force on block A = gradient $= \frac{-2.00 - (-10.00)}{3.00 \times 10^{-3}}$ $= 2670 \text{ N (3 s.f.)}$ Force on block A = – Force on block C $= -2670 \text{ N}$
(b)(ii)	$v_A = \frac{p_A}{m_A}$ $= \frac{-2.00}{5.0}$ $= -0.40 \text{ m s}^{-1}$
(b)(iii)	Total momentum conserved: $p_i = p_f$ $p_i = p_{fA} + p_{fC}$ $p_{fC} = p_i - p_{fA} = -10.00 - (-2.00) = -8.00 \text{ N s}$ $v_C = -\frac{8.00}{10} = -0.80 \text{ m s}^{-1}$

	Total KE before collision = $\frac{1}{2}m_A u^2$ $= \frac{1}{2}(5.0)\left(\frac{-10.00}{5.0}\right)^2$ $= 10 \text{ J}$ Total KE after collision = $\frac{1}{2}[m_A v_A^2 + m_C v_C^2]$ $= \frac{1}{2}[(5.0)(-0.40)^2 + (10)(-0.80)^2]$ $= 3.6 \text{ J}$ <u>OR</u> relative speed of approach = $u_A - u_C$ $= 2.0 \text{ m s}^{-1}$ relative speed of separation = $v_C - v_A$ $= 0.80 - 0.40$ $= 0.40 \text{ m s}^{-1}$
	Since total KE is not conserved, the collision is not elastic. OR Since relative speed of approach is not equal to the relative speed of separation, the collision is not elastic.
Q3	